

# Requirements and service boundaries

Project: Grounded Learning Service

## 1. Your mission

### Your learning target

I can explain the mechanism and terminology of requirements and service boundaries.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Neural network experiment defense. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of requirements and service boundaries

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Retrieve what you already know

Recall AI 5 exit evidence, tested Python programs, APIs, structured data, version control and basic system diagrams.. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

#### My starting explanation

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## This week's end goal

Explain requirements and service boundaries using the deciding step in this fictional service.

## 2. Notice the evidence

A service boundary states who can request what, what data may be read and which actions are outside scope. Requirements must be observable.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

**List the given facts and one fact that is not supplied.**

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## 3. See how it works

The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.

### Requirements and service boundaries — evidence trace

**Input and scope · Check or calculation · Conclusion and limit**

**Input and scope:** A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00.

**Check or calculation:** The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.

**Conclusion and limit:** Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the service boundary.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the

stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

### Compare a familiar case with a changed case

A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

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### Change one thing in the pictured case

Change: Replace P with Q, which has no approved card.

Prediction: The year is unsupported, and the ticket sale is still excluded.

Reason: Knowledge and permission are separate requirements.

## 5. Words that unlock the idea

### Word · Meaning

**Word:** Service boundary

**Meaning:** The declared inputs, data and operations of a service

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**Word:** Acceptance criterion

**Meaning:** An observable requirement

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**Word:** Out of scope

**Meaning:** Not authorised by this service contract

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**Use two terms to explain a step in this week's example.**

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## 6. Follow a worked example

A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00.

Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the service boundary.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Requirements and service boundaries

Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket.

### The facts we will use

Approved card P says 1912. The contract permits reading cards, not selling tickets.

## Requirements and service boundaries

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Read approved card

The supplied year is 1912.



#### 2. Split the request

Answering the year and selling a ticket are separate operations.



#### 3. Apply the boundary

Return 1912 with card P; do not sell a ticket.

### CHANGE ONE THING

Replace P with Q, which has no approved card.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Read approved card

**What happens and why:** The supplied year is 1912.

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**Step:** 2. Split the request

**What happens and why:** Answering the year and selling a ticket are separate operations.

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### Step: 3. Apply the boundary

**What happens and why:** Return 1912 with card P; do not sell a ticket.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

A service boundary states permitted work; useful information does not grant extra operations.

### What this example does not establish

An approved card can be outdated; this trace does not independently verify museum history.

## 7. Find and repair a mistake

A helpful answer does not grant an operation that the service contract excludes.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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### Locate the earliest unsupported step

A report omits this qualification: A helpful answer does not grant an operation that the service contract excludes. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**State one repair and the checks you would rerun.**

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## 8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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**Explain your prediction to a partner before comparing results.**

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### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Requirements and service boundaries”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

#### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case.

Each person writes their own explanation.

### Check your reasoning

Use the worked case for Requirements and service boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Split the museum visitor’s request into two operations. Which can the service perform for P, and what happens to each operation when Q has no approved card?

#### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

### Predict, investigate and preserve the evidence

A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00. Show the intermediate steps and give the conclusion with its scope.

#### Record · Your evidence

**Record:** Given input and fixed rule

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Prediction before checking

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Actual result, or “not run”

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** First mismatch and one explanation

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Boundary or missing input and expected response

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

### Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

### Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: A helpful answer does not grant an operation that the service contract excludes. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

**Explain which evidence supports your conclusion and which claim remains unproved.**

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## 11. Connect your project

Add a labelled requirements and service boundaries evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of requirements and service boundaries. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### Project evidence · Record

**Project evidence:** Component and version

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Inputs permitted at prediction time

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Trace and comparison with earlier work

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Failure, limit and human review

**Record:** \_\_\_\_\_

\_\_\_\_\_

### Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

## 12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

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For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

#### **My limited claim and a useful next check**

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## 13. Study a complete case

### Complete evidence record

#### Requirements and service boundaries — evidence trace

##### **Input and scope · Check or calculation · Conclusion and limit**

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

##### **My complete trace of the worked case**

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# 14. Try a new case independently

## Independent case: Apply a new case independently

A read-only inventory service is asked to list item A and delete item B. Inventory says A is available. Trace permitted work.

**My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## Attempt this without hints

A read-only inventory service is asked to list item A and delete item B. Inventory says A is available. Trace permitted work.

**My independent procedure, result and limitation**

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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

**Use feedback to improve**

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.